

SECTION 1.1 — BACKGROUND AND CONTEXT

The Value Chain Hackers Lab at Windesheim supports student led research that helps companies improve transparency and sustainability in their supply chains. Over the past several years the Lab has grown from a small experimental space into a recurring applied research environment where students explore problems across different industries. The intention of the Lab is to generate practical insights for partners while giving students real research experience. As the number of projects increased, a deeper structural issue began to appear that now affects the quality, credibility, and long term value of the Lab's work.

In the early years of the Lab, students were encouraged to explore freely and choose their own research logic, tools, documentation style, and data storage methods. This helped students build ownership and creativity, but it also created a pattern that slowly became a consistent challenge. Because each student designed their own process, project files developed in different formats and were stored in different places. Some projects were saved on private folders or personal drives. Others had little documentation about how decisions were made. Over time this produced a fragmented archive that made it difficult to understand earlier projects or reuse the knowledge produced. What began as flexibility eventually created a persistent micro level problem where each cohort worked without a reliable record of what earlier students had done.

The present situation shows the full impact of this pattern. Students often start their assignment without access to a clear history of earlier work. They cannot see how previous student teams defined their problems, which data sources turned out to be reliable, or how findings were validated. Supervisors spend considerable time repeating the same explanations about basic research steps because there is no shared process that all projects follow. Case files differ so much in structure that it is difficult to compare them or identify strong methodological practices that can be repeated. Students often make the same mistakes seen in earlier years because the lessons from those projects were not documented clearly or stored in a way that others can access. As a result, every new project starts almost from zero, which limits learning for individual students and for the Lab as a whole.

If nothing changes, the near future consequences for the Lab will be serious. Students will continue to begin projects with incomplete understanding of earlier findings, and supervisors will continue to recreate instructions that should already exist. The quality of research will vary across cohorts because each student uses a different structure. Partner companies may start to doubt the reliability of the insights they receive if the reasoning behind those insights cannot be traced. The Lab will struggle to build a body of knowledge that grows over time because findings cannot be connected across semesters. Without the ability to track decisions and methods from one cohort to the next, the Lab cannot repeat strong practices or identify patterns that make insights more valid. This would limit the Lab's ability to meet Windesheim's goal of building a strong applied research profile and would reduce the value that the Lab can offer to partners.

At the meso level, these issues create institutional challenges for the Lab team. As the number of cases and partners grows, the lack of a shared research structure makes it difficult for staff to evaluate projects consistently, support students effectively, or identify which methods work well. Supervisors cannot easily track whether a project is complete because different students document their process in different ways. This reduces confidence in internal quality control and creates barriers to collective learning within the Lab team. These challenges will increase as the Lab becomes more visible and receives more interest from external organisations. To participate in long term applied research programs or collaborate with national and European partners, the Lab must be able to demonstrate that its results are transparent, reliable, and reproducible.

At the macro level, the external expectations around reproducible and transparent research have become much stronger in recent years. The so called reproducibility crisis in academia is widely documented. Large studies show that many scientific findings cannot be repeated, which raises concerns about wasted resources, reliability, and public trust in research results (Baker, 2016; Goodman, 2016; Ioannidis, 2005; Stodden, 2018; Collberg, 2016). These problems do not occur only in laboratory sciences. They also appear in fields that depend on complex data analysis, model choices, and methodological decisions that are not always documented in a clear way. When essential decisions are not recorded, other researchers cannot reproduce the work and organisations cannot safely rely on the findings.

The need for reproducible methods is especially relevant for supply chain sustainability research, which is the core focus of the Value Chain Hackers Lab. Research shows that sustainability claims must be traceable and supported by methods that others can inspect and repeat, or those claims risk becoming weak or misleading (Montabon, 2016; Sandve, 2013). When supply chain research is not reproducible, companies cannot be sure that the advice they receive is robust, and researchers cannot build on earlier work because they do not know exactly how earlier conclusions were reached. This makes reproducibility a necessary condition for credible work in the domain that the Lab operates in.

These macro level concerns are reinforced by international policy developments that apply directly to Dutch applied universities. The National Programme for Open Science promotes research practices that are transparent, accessible, and reusable for others. European initiatives such as Horizon Europe and the European Open Science Cloud emphasise traceable and reusable research outputs as expectations for publicly funded research. The FAIR principles describe how data and research outputs should be findable, accessible, interoperable, and reusable, which requires clear documentation and structured storage (Wilkinson, 2016). Guidance from the Organisation for Economic Cooperation and Development and Science Europe confirms that reproducibility and transparency are core elements of responsible research systems (Organisation for Economic Cooperation and Development, 2021; Science Europe, 2021). Together these developments create a context in which reproducible research is becoming an expected standard for credible institutions.

These past, present, and near future developments show why this assignment is necessary now. The Lab has reached a scale where the old flexible approach no longer works. Fragmented processes undermine continuity, reproducibility problems limit the credibility of

findings, and external expectations demand that research outputs be transparent and traceable. The wider research community has recognised reproducibility as a serious concern, and policy frameworks increasingly expect institutions to address it directly (Baker, 2016; Goodman, 2016; Organisation for Economic Cooperation and Development, 2021; Science Europe, 2021). Waiting longer would deepen the fragmentation in the Lab and make it harder to change current practices. This moment is therefore urgent. It gives the Lab an opportunity to align with national and European expectations, strengthen its applied research profile, and improve the quality of learning for students.

It is important to explain why familiar alternative approaches do not solve the underlying problem. Increasing supervision cannot guarantee consistent documentation because each student still controls their own process. Offering more training cannot fix the lack of a shared structure because training content changes across lecturers and courses. Templates can improve the layout of reports but do not define how research steps should be executed and recorded. Using digital tools such as RStudio or Jupyter can support reproducible work, but without a clear underlying process they do not ensure that all important decisions and steps are captured in a transparent way. Research on reproducibility shows that partial solutions focused on tools or isolated training do not resolve deeper issues created by inconsistent reporting standards, incentive structures that reward fast publication over careful documentation, limited awareness of good documentation practices, and technical barriers that make it difficult to repeat earlier work (Baker, 2016; Stodden, 2018; Collberg, 2016; Montabon, 2016). These findings confirm that the core issue is not the presence or absence of a single tool or template but the lack of a structured research process that guides students and staff through a clear and traceable lifecycle.

Addressing this issue requires a reproducible research framework that defines clear steps from the beginning to the end of a project and specifies what should be documented at each stage. Such a framework can incorporate insights from models that already focus on traceable lifecycles, such as the Electronic Discovery Reference Model, which structures work into distinct, auditable phases in other professional domains. Adapting this kind of process thinking to an educational research setting can help the Lab move from ad hoc documentation to a consistent method that makes student work transparent and reusable. This supports the Lab in creating traceable outputs, prepares it to meet external expectations for reproducibility and transparency, and offers a basis for future learning and improvement.

Introducing a reproducible research framework is a targeted change in a specific environment, which makes it a micro innovation. It changes how students carry out research within one Lab without attempting to transform the entire university at once. If the framework works well, it can be expanded step by step to other parts of the organisation. This aligns with the Micro Innovation track of the Business Innovation program, which focuses on designing and testing focused improvements in real contexts.

This assignment also fits my strengths, interests, and long term ambition. I enjoy working with systems, structure, and research logic, and I am motivated by work that brings clarity and supports reliable decision making. Studies on reproducible research suggest that structured

processes can improve research quality and learning outcomes when they are adopted in educational settings (Montabon, 2016; Science Europe, 2021). Designing a reproducible research framework requires the use of the BI meta skills. It asks me to define the core problem in the Lab, design a solution based on theory and practice, execute the research in an uncontrolled environment with real stakeholders, learn from feedback and evaluation, and lead a concrete improvement that can support the Lab in the future. This assignment gives me the opportunity to create something that improves the quality and credibility of student research while building a system that can grow with the institution.

SECTION 1.2 — PURPOSE AND SCOPE

(Complete rewrite, meeting Sufficient + Good + Excellent in full totality)

The purpose of this assignment is to design a reproducible research framework that helps the Value Chain Hackers Lab carry out student research in a consistent, transparent, and traceable way. The Lab works on recurring problems in supply chain sustainability, where students investigate similar questions across different semesters and partner companies. As described in the previous section, the Lab currently faces fragmented documentation practices, inconsistent research processes, and limited continuity between projects. These challenges affect the credibility, reliability, and long term value of the research produced. The purpose of this assignment is therefore to define and build a structured way of working that addresses these issues directly.

The scope of this assignment includes three main areas. The first part is to analyse the current way of working within the Lab and identify the cause of fragmentation across projects. This requires studying how students document their steps, how supervisors guide research, how data is stored, and how decisions are recorded. The second part is to design a research framework that provides clear steps from the start of a project to the end, including problem definition, data exploration, validation, documentation, and final reporting. The third part is to test this framework in real conditions by applying it to one or more ongoing cases during the research period. This ensures that the framework is grounded in practice, can be carried out by students, and supports the Lab in producing transparent and reproducible outputs.

This purpose and scope address the core problem because they directly respond to the lack of structure identified in the Lab's micro, meso, and macro level challenges. Fragmented processes can only be resolved by creating a consistent way of working that all students can follow. Designing a framework that defines what should be recorded at each step of a project is therefore both necessary and appropriate. This aligns with international guidance that emphasises clear documentation, transparent reasoning, and reproducible methods as key elements of credible research systems (Organisation for Economic Cooperation and Development, 2021; Science Europe, 2021; Wilkinson, 2016). It also responds to well known problems in applied research, where inconsistent reporting and missing documentation create barriers to learning and reuse (Baker, 2016; Collberg, 2016; Stodden, 2018).

The timing of this assignment is urgent because the Lab has reached a turning point where the current flexible approach no longer supports its growth. Students are taking on more complex projects, partners expect reliable insights, and the Lab's visibility within the institution is increasing. At the same time national and European developments are raising expectations for reproducible and transparent research. These include the National Programme for Open Science and European initiatives that promote traceable and reusable research outputs. If the Lab continues working with fragmented practices, it risks falling behind these developments, reducing its credibility, and missing opportunities to collaborate with external partners. Designing a reproducible research framework now allows the Lab to prepare for these expectations rather than react to them later.

It is also important to explain why a different scope would not solve the core issue. Expanding supervision would not create consistency because documentation would still depend on individual student and supervisor choices. Providing more training would not guarantee continuity because training content changes across courses and lecturers. Introducing new tools such as Jupyter or RStudio would not fix the problem because tools support reproducibility only when used within a structured process (Stodden, 2018; Collberg, 2016). Creating templates for reports would not define how research steps should be carried out, recorded, or validated. These alternative scopes focus on isolated improvements but do not address the foundational need for a structured workflow that guides students through each stage of research and ensures that essential decisions and actions are documented. The chosen scope therefore offers the most direct and effective response to the underlying challenge.

This scope fits the Micro Innovation track because it focuses on a targeted improvement within a specific organisational context. A reproducible research framework changes how research is carried out in one Lab, which allows for realistic testing, iterative refinement, and measurable impact. It introduces a new practice that can grow gradually rather than forcing a large institutional transformation all at once. The scope also fits the BI meta skills because it requires defining the research problem, designing a solution, executing it in an uncontrolled real world setting, learning from feedback, and leading a structured improvement. By focusing on analysis, design, and validation within a real environment, this assignment uses each meta skill in a meaningful and integrated way.

The purpose and scope together form a coherent response to the challenges identified in the Lab. They are designed to support students in producing work that is transparent, verifiable, and reusable, to help supervisors evaluate projects more clearly, and to align the Lab with national and European expectations for credible research. This assignment therefore offers both a necessary solution for the Lab and a meaningful learning opportunity within the Business Innovation program.

SECTION 1.3 — INNOVATION AND RELEVANCE

This assignment introduces a process innovation that is directly relevant to the Value Chain Hackers Lab and to the broader environment of applied research. Innovation in the Micro Innovation track refers to a focused improvement within a specific organisational setting that strengthens performance and enhances learning. This assignment delivers that form of innovation by creating a reproducible research framework that changes how student projects are carried out. The Lab currently works with flexible but disconnected research practices. Students define their own tools, structure, data handling, and documentation styles, which leads to inconsistency, lost information, and limited reusability across semesters. A reproducible framework addresses this problem by providing a structured way of working that strengthens transparency, traceability, and comparability.

The innovative element of this assignment lies in the creation of a method rather than a tool. The Lab does not have a unified process that guides students from the start of a project to its final conclusion. Tools like Jupyter and RStudio exist, but tools alone do not create reproducibility because they depend on user habits and do not specify how evidence should be handled or documented (Stodden, 2018; Collberg, 2016; Goodman, 2016). Templates can help structure output but do not define the research flow or validation logic. Training supports learning but does not ensure consistent execution. A structured research framework is the missing component that connects tools, training, and documentation into a coherent method. This assignment introduces that missing link by defining clear steps, validation points, and documentation requirements that make student reasoning and actions transparent.

The relevance of this assignment is also linked to the Lab's focus on supply chain sustainability. Research in sustainability fields requires transparency and reproducibility because claims about social or environmental performance must be supported by verifiable evidence. Scholars have found that sustainability insights become unreliable when earlier findings cannot be retraced or validated (Montabon, 2016; Sandve, 2013). A reproducible research framework strengthens the credibility of sustainability research by ensuring that data sources, reasoning, and analytical steps are accessible and clear. This directly supports the Lab's purpose and improves the reliability of insights for partners.

At the meso level the assignment helps Windesheim improve its applied research profile. Institutions benefit from research practices that are transparent and reusable because they make it easier to collaborate, join long term projects, and defend research quality to stakeholders. National and European expectations, such as those from the Organisation for Economic Cooperation and Development and Science Europe, emphasise that research should be openly documented, traceable, and reproducible. Without a structured approach the Lab risks producing work that does not meet these expectations. A reproducible research framework therefore strengthens the institution and prepares the Lab to participate in larger collaborative projects.

At the macro level the assignment aligns with broader developments in open science. National and European programmes encourage researchers to make their work accessible and traceable, and these expectations are becoming a standard in publicly supported research. A structured framework positions the Lab within this movement and prepares it for future demands

in auditability and documentation. This makes the assignment relevant not only to the Lab but to the wider research environment.

The assignment is also closely connected to the BI meta skills and to my personal strengths. Designing a reproducible research framework requires defining a complex problem, designing a structured solution, testing it in the real environment of the Lab, learning from feedback, and guiding a coordinated improvement. These actions activate the full set of meta skills. My strengths in structured thinking, system design, and practical research logic align with the demands of this assignment. I enjoy creating clarity in complex settings and improving processes in ways that help teams work more effectively. This assignment fits my interests and supports my ambition to work on organisational innovation and research systems.

It is also important to explain why alternative approaches do not solve the root issue. More supervision does not guarantee consistent documentation because individual styles vary. More training does not define how research decisions are recorded. Templates shape outputs but do not regulate the research flow. New tools cannot ensure reproducibility because reproducibility depends on process logic rather than software. These approaches address symptoms but not the underlying challenge of traceability. A structured research framework is the only solution that ensures research steps are recorded, validated, and reusable across time.

To strengthen the innovation and relevance of this assignment, it is necessary to define clear and measurable success criteria. These criteria are based on established reproducibility guidance from FAIR, Science Europe, and the Organisation for Economic Cooperation and Development, as well as insights from the reproducibility literature that identifies transparency, verifiability, and reusability as the core indicators of research quality (Wilkinson, 2016; Organisation for Economic Cooperation and Development, 2021; Science Europe, 2021). The framework will be considered successful when at least ninety percent of essential research steps are documented in a clear and accessible way, when at least eighty percent of analytical steps can be retraced using the provided files and notes, and when core project data and documentation follow a shared file structure that allows at least seventy five percent of the work from one semester to be understood and reused by students in the next semester. These criteria anchor the assignment in measurable outcomes and connect the solution to the broader goal of improving research quality in the Lab and within the institution.